(A) Task schematic

Example Context Identification trial

Inputs:

$$X_q = \begin{bmatrix} 0 & 0 & \mathbf{1} & 0 & 0 \end{bmatrix} \quad \text{(reference: feature 2)}$$

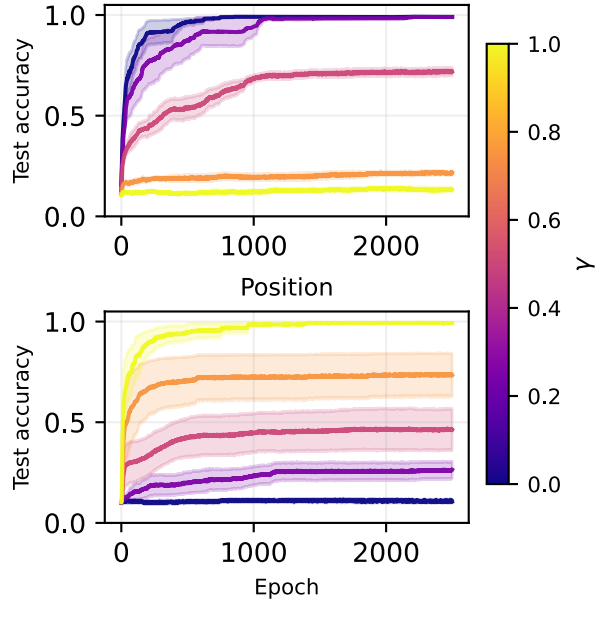
$$X_c = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{array}{l} \leftarrow \text{feature 4, dist=2} \\ \leftarrow \text{feature 0, dist=2} \\ \leftarrow \text{feature 0, dist=2} \\ \leftarrow \textbf{feature 1, dist=1} \\ \leftarrow \text{feature 4, dist=2} \end{array}$$

Targets:

$$y_{\text{feature}} = \begin{bmatrix} 0 & \mathbf{1} & 0 & 0 & 0 \end{bmatrix} \quad \text{(feature index 1)}$$

$$y_{\text{position}} = \begin{bmatrix} 0 & 0 & 0 & \mathbf{1} & 0 \end{bmatrix} \quad \text{(context position 3)}$$

(B) Fixed γ sweep



(C) Learnable γ

